

# EEE MSM Predictions v2

## MSM Population Estimation and Simulation Methodology

October 2025

### Abbreviations

- **MSM**: Men who have sex with men
- **SSM**: Same-sex male households
- **FF**: Female-female same-sex households
- **MM**: Male-male same-sex households
- **SS**: Total same-sex households
- **HH**: Total households
- **GSS**: General Social Survey
- **ACS**: American Community Survey
- **NCHS**: National Center for Health Statistics

### County-Level MSM Population Estimation

We estimate the population of men who have sex with men (MSM) at the county level using a multi-step approach that incorporates household structure, urbanicity stratification, and demographic weighting.

#### 1. Same-Sex Male Household Imputation

We estimate the number of same-sex male (SSM) households in each county  $c$  as:

$$\widehat{SSM}_c = SS_c \cdot \left( \frac{MM_c}{MM_c + FF_c} \right)$$

where:

- $SS_c$ : Total same-sex households in county  $c$
- $MM_c$ : Estimated male-male households
- $FF_c$ : Estimated female-female households

## 2. MSM Household Index

We compute a household-based index to scale county-level estimates relative to urbanicity-wide norms:

$$\text{Index}_c = \frac{\widehat{SSM}_c / HH_c}{\overline{SSM}_u / \overline{HH}_u}$$

where:

- $HH_c$ : Total households in county  $c$
- $u$ : Urbanicity classification
- $\overline{SSM}_u, \overline{HH}_u$ : Aggregated SSM and total households for urbanicity  $u$

## 3. Base MSM Proportion by Urbanicity

From the General Social Survey (GSS), we extract urbanicity-stratified base MSM proportions:

$$P_u = \mathbb{P}(\text{MSM} \mid u)$$

## 4. County-Level MSM Estimate

We estimate the MSM population in county  $c$  as:

$$\widehat{M}_c = P_u \cdot \text{MalePop}_c \cdot \text{Index}_c$$

where  $\text{MalePop}_c$  is the estimated male population aged 18+ in county  $c$ .

## 5. Demographic Breakdown (Revised)

For each demographic subgroup  $d$  (e.g., age, education, income), we compute:

1. Raw estimated MSM subgroup counts:

$$\widehat{M}_{c,d} = p_{u,d} \cdot \text{MalePop}_c$$

2. Normalized subgroup shares:

$$s_{c,d} = \frac{\widehat{M}_{c,d}}{\sum_{d'} \widehat{M}_{c,d'}}$$

3. Final scaled MSM estimates per subgroup:

$$\widehat{M}_{c,d} = s_{c,d} \cdot \widehat{M}_c$$

## 6. Simulation Method

| Quantity Simulated                              | Distribution Used                                              |
|-------------------------------------------------|----------------------------------------------------------------|
| Male population (county-level)                  | Normal (mean from ACS, SE from MOE/1.645)                      |
| Male population (age, education, income groups) | Normal (group-specific estimates, with SE from MOE)            |
| Total households (county-level)                 | Normal (with SE from MOE)                                      |
| Same-sex households (MM, FF, SS)                | Log-normal (after log-transform of mean and SE)                |
| MSM proportion (by urbanicity)                  | Logit-normal                                                   |
| MSM proportion (by demo $\times$ urbanicity)    | Logit-normal                                                   |
| MM / (MM + FF) ratio                            | Logit-normal (logit transformed ratio, SE derived from counts) |

**Table 1:** Simulated input quantities and assumed distributions for uncertainty propagation

We use the following transformation to simulate draws from a logit-normal distribution:

$$\mu = \text{logit}(p), \quad \text{SE}_{\text{logit}} = \frac{\text{SE}}{p(1-p)}, \quad p^{(\text{draw})} = \text{logit}^{-1}(\mathcal{N}(\mu, \text{SE}_{\text{logit}}))$$

## 7. Data Sources

This methodology relies on multiple data sources and assumptions. These include:

| Data Source                                | Variable(s)                                                                                                                                                                                     | Usage in MSM Estimation                                                                                                            |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>ACS 2022</b>                            | <ul style="list-style-type: none"> <li>- Total households (<math>HH_c</math>)</li> <li>- Male population 18+ (<math>MalePop_c</math>)</li> </ul>                                                | <ul style="list-style-type: none"> <li>- Denominator for SSM index</li> <li>- Base population for MSM count calculation</li> </ul> |
| <b>ACS 2023 (Age, Educ), 2021 (Income)</b> | <ul style="list-style-type: none"> <li>- Demographic distributions: age, education, income</li> </ul>                                                                                           | <ul style="list-style-type: none"> <li>- Weighting male population by subgroup for demographic breakdown</li> </ul>                |
| <b>ACS 2014</b>                            | <ul style="list-style-type: none"> <li>- Same-sex male households (<math>MM_c</math>),</li> <li>- Female same-sex (<math>FF_c</math>),</li> <li>- Total same-sex (<math>SS_c</math>)</li> </ul> | <ul style="list-style-type: none"> <li>- Used to compute MM/SS ratio</li> <li>- Derivation of SSM index</li> </ul>                 |
| <b>ACS 2022 (for SS)</b>                   | <ul style="list-style-type: none"> <li>- Same-sex households (SS)</li> </ul>                                                                                                                    | <ul style="list-style-type: none"> <li>- Used to update total SS counts by county</li> </ul>                                       |
| <b>GSS 2016–2024</b>                       | <ul style="list-style-type: none"> <li>- MSM prevalence by urbanicity (<math>P_u</math>)</li> <li>- MSM prevalence by subgroup and urbanicity (<math>p_{u,d}</math>)</li> </ul>                 | <ul style="list-style-type: none"> <li>- Determines baseline and demographic-specific MSM rates</li> </ul>                         |
| <b>NCHS Urban-Rural Classification</b>     | <ul style="list-style-type: none"> <li>- Urbanicity designation (<math>u</math>) for each county</li> </ul>                                                                                     | <ul style="list-style-type: none"> <li>- Stratifies counties for applying appropriate MSM proportions</li> </ul>                   |

**Table 2:** Data sources, variables, and their roles in MSM population estimation